COURSE SLIDES · WSQ

QUANTUM

# Securing the Future with Quantum Computing and Cryptography

WSQ Course Code: TGS-2024043419

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version 11.0 · 11 August 2026

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COURSE ADMINISTRATION

# Welcome & Housekeeping


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# Digital Attendance (Mandatory)

1. Trainer displays the digital attendance QR code from the SSG portal
2. Scan the QR code with your mobile phone camera
3. Submit your attendance — AM, PM and Assessment, every day
4. 75% minimum attendance is required for assessment and funding


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# About the Trainer




## Your Trainer

General Trainer template —
to be completed by the trainer

**NAME**

**TITLE / DESIGNATION**

**QUALIFICATIONS**

**AREAS OF EXPERTISE**

**TRAINING & INDUSTRY EXPERIENCE**

**CONTACT**

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YOUR TRAINER

# About the Trainer




## Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

**ROLE**
Principal Trainer, Tertiary Infotech Academy Pte Ltd

**QUALIFICATIONS**
PhD; ACTA / DACE certified adult educator.

**EXPERTISE**
Quantum computing, quantum cryptography, data science and AI engineering.

**DELIVERS**
WSQ courses on quantum computing, cybersecurity and applied machine learning.

**FOUNDER**
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.


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ICE-BREAKER

# Let's Know Each Other

- Your name, organisation and role.
- Your background in physics, mathematics or programming (if any).
- What you want to be able to do with quantum computing after this course.


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HOUSEKEEPING

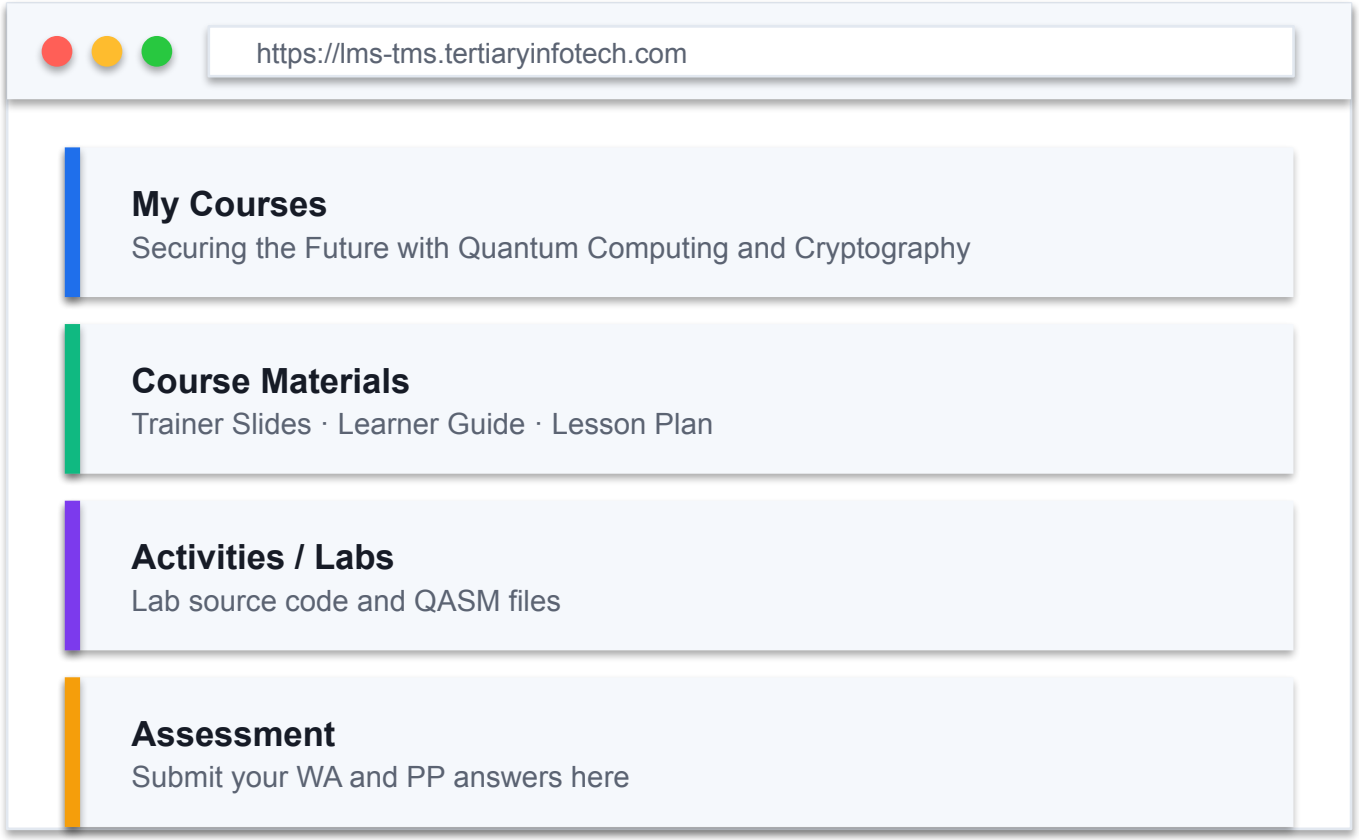
# Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required for funding.


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COURSE PORTAL · LMS/TMS

# Download Your Course Material




1. Go to lms-tms.tertiaryinfotech.com and sign in with the account you were issued.
2. Open My Courses and select this course.
3. Download the Trainer Slides, Learner Guide and Lesson Plan.
4. Keep them open — the assessment is OPEN BOOK.

**Download the slides and Learner Guide BEFORE the assessment — they are your permitted open-book materials.**


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WHAT YOU'LL ACHIEVE

# Learning Outcomes

1

## LO1 · Mathematical models

Design the mathematical models required for quantum computing applications.

2

## LO2 · Physical methodologies

Decide on appropriate physical methodologies for quantum computing problems.

3

## LO3 · Two-level models

Direct two-level quantum models for quantum computing.

4

## LO4 · Business applicability

Evaluate quantum computing applicability in the business context.

5

## LO5 · Quantum gates

Spearhead the application of quantum gates to quantum computing domains.

6

## LO6 · Algorithm selection

Establish guidelines for selecting quantum computing algorithms.

7

## LO7 · Synthesis

Synthesize quantum methodologies to address complex quantum computing problems.


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SCHEDULE

# Lesson Plan — 4 Days, 8 hours/day

## Days 1–2

- Day 1 — Quantum Physics Fundamentals & Quantum Computing Systems
  - Topic 1: Quantum Physics Fundamentals (Activity 1)
  - Topic 2: Quantum Computing Systems (Activities 2–3)
- Day 2 — Quantum Gates, Circuits and Quantum Protocols
  - Topic 3: Quantum Gates and Circuits (Activities 4–5)
  - Topic 4: Quantum Protocols (Activities 6–8)

## Days 3–4

- Day 3 — Quantum Algorithms and Quantum Cryptography
  - Topic 5: Quantum Algorithms (Activities 9–12)
  - Topic 6: Quantum Cryptography & Safety (Activities 13–14)
- Day 4 — Applications, Error Correction and Assessment
  - Topic 7: Applications (Activities 15–16)
  - Topic 8: Quantum Error Correction (Activities 17–18)
  - Final Assessment: WA (1 hr) + PP (1 hr)


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BEFORE YOU BEGIN

# Briefing for Assessment

1

## Phones away

Place phones and other materials under the table or on the floor.

2

## No recording

No photos or recording of assessment scripts.

3

## No discussion

Work individually — no discussion during the assessment.

4

## Black or blue pen

Use a black/blue pen for hard-copy assessments.

5

## No correction fluid

No liquid paper or correction tape on any script.

6

## Time is called

Scripts are collected when the time is up.


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FINAL ASSESSMENT

# Assessment

## Written Assessment (WA)

- Short-Answer Questions (SAQ)
- 1 hour · open book
- Tests underpinning KNOWLEDGE
- Covers all eight topics

## Practical Performance (PP)

- Hands-on quantum tasks
- 1 hour · open book
- Tests PRACTICAL ability
- Built on the class activities

**OPEN BOOK — course slides, Learner Guide and approved materials only**

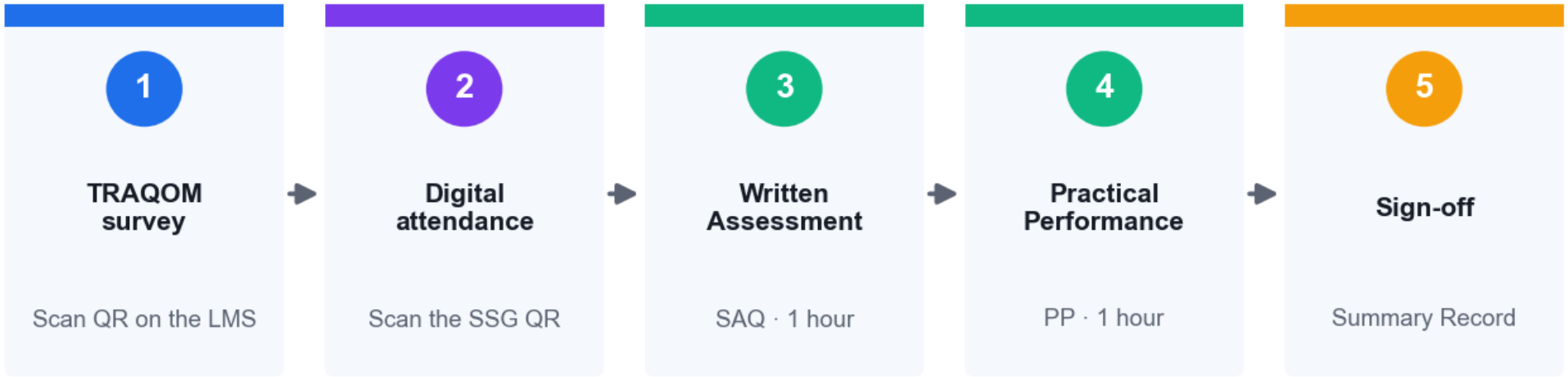
**Eligibility:** minimum 75% attendance + assessed Competent. An appeal process is available.


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ON ASSESSMENT DAY

# Assessment Flow

## What happens on assessment day




**Open book — course slides, Learner Guide and approved materials only**

Complete every step in order — the TRAQOM survey and digital attendance are mandatory for funding.


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GETTING STARTED

# Why Quantum, Why Now


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THE CORE IDEA

# What Makes a Qubit Different

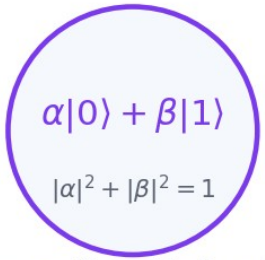
**CLASSICAL BIT**

0 1

Exactly one value at a time

**n bits store 1 of $2^n$ values**

**QUBIT**




Superposition of both values

**n qubits hold all $2^n$ amplitudes**

**Classical bit**

Holds exactly one value — 0 or 1 — at any instant.

**Qubit**

Holds a superposition of both, described by two complex amplitudes.

**Measurement**

Collapses the superposition to a single classical outcome.

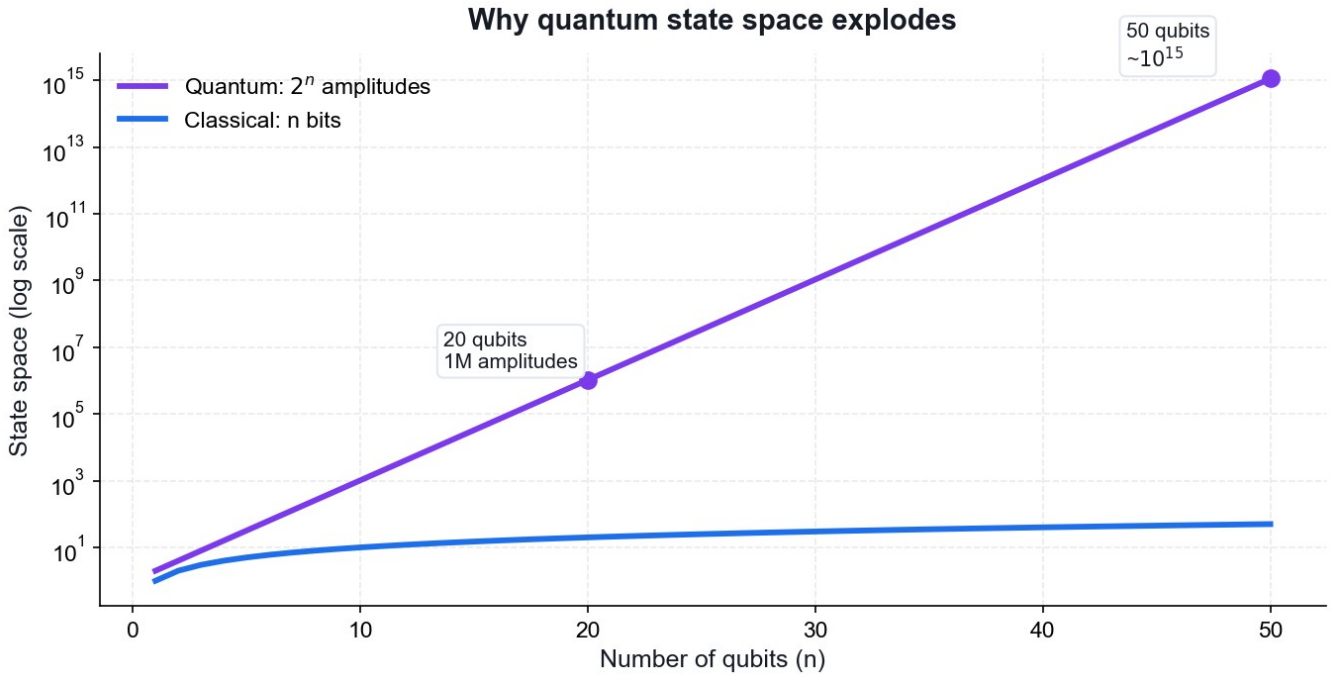
**The payoff**

n qubits carry $2^n$ amplitudes at once — quantum parallelism.


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WHY IT MATTERS

# The State Space Explosion




**50 qubits**

~$10^{15}$ amplitudes — beyond any classical supercomputer's memory.

**300 qubits**

More amplitudes than atoms in the observable universe.

**But**

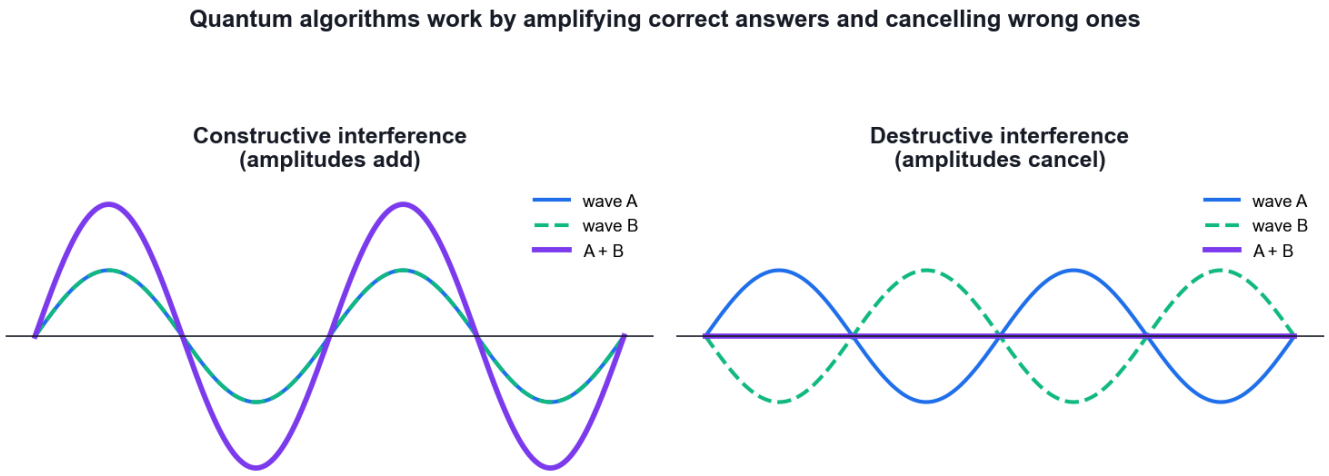
You only ever read out n classical bits — algorithms must be clever.

**So**

Interference, not brute force, is what gives quantum its advantage.


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# Interference — The Real Engine




**Amplitudes can cancel**

Unlike probabilities, amplitudes are complex and can be negative.

**Wrong answers cancel**

A good algorithm arranges destructive interference on wrong paths.

**Right answers add**

Constructive interference concentrates probability on the answer.

**Every algorithm**

Deutsch, Grover and Shor are all interference engines.


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THE MENTAL MODEL

# Superposition supplies the paths. Interference picks the answer.

Every quantum algorithm in this course follows that pattern — prepare a superposition, apply an oracle, then interfere so the correct answer survives.


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01

TOPIC 01

# Quantum Physics Fundamentals

Postulates of Quantum Mechanics · Mathematics for Quantum Mechanics


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MAPS TO LO1

# Key Concepts — Quantum Physics Fundamentals

1

## The five postulates

State space, unitary time evolution, observables, the Born rule and wave-function collapse are the axioms every quantum algorithm obeys.

2

## Hilbert space

Quantum states are unit vectors in a complex inner-product space; a single qubit lives in a 2-dimensional Hilbert space.

3

## Unitary operators

U†U = I preserves the norm — this is why every quantum gate is reversible and probability is conserved.

4

## Hermitian operators

A† = A guarantees real eigenvalues, so Hermitian operators represent physically measurable observables.

5

## Dirac notation

Bra-ket notation compactly expresses states $|\psi\rangle$, dual vectors $\langle\psi|$, inner products $\langle\varphi|\psi\rangle$ and outer products $|\psi\rangle\langle\varphi|$.

6

## Tensor products

The state space of a multi-qubit system is the tensor product of the single-qubit spaces — dimension grows as $2^n$.


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THE THREE OBJECTS YOU NEED

# The Mathematical Toolkit

## State vector

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

A qubit is a unit vector in a 2-dimensional Hilbert space, with $|\alpha|^2 + |\beta|^2 = 1$.

## Unitary operator

$$U^\dagger U = I$$

Every quantum gate is unitary — reversible, and probability-preserving.

## Hermitian operator

$$A^\dagger = A$$

Observables are Hermitian, which guarantees real measured eigenvalues.


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BRA-KET NOTATION

# Dirac Notation at a Glance

## Ket / Bra

$|\psi\rangle$ and $\langle\psi|$

A ket is a column vector; a bra is its conjugate transpose (row vector).

## Inner product

$\langle\varphi|\psi\rangle$

A single complex number — the overlap, or probability amplitude, between two states.

## Born rule

$P(x) = |\langle x|\psi\rangle|^2$

The probability of measuring outcome x is the squared magnitude of the amplitude.


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# The Five Postulates

| # | Postulate | What it says |
|---|---|---|
| 1 | State space | A system's state is a unit vector in a complex Hilbert space. |
| 2 | Time evolution | A closed system evolves by a unitary operator: $\|\psi'\rangle = U\|\psi\rangle$. |
| 3 | Observables | Measurable quantities are Hermitian operators with real eigenvalues. |
| 4 | Born rule | The probability of an outcome is the squared amplitude. |
| 5 | Collapse | After measurement the state collapses to the measured eigenstate. |


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WHAT YOU'LL DO

# Hands-On Activities — Quantum Physics Fundamentals

## Activity 1

- Linear Algebra for Quantum Mechanics in Python

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TOPIC 01 · HANDS-ON ACTIVITY

ACTIVITY 1

# Linear Algebra for Quantum Mechanics in Python

Work with the mathematical objects every quantum algorithm is built from: state vectors, inner and outer products, unitary and Hermitian operators, eigenvalues and tensor products — all in NumPy.

**MAPS TO** LO1 — Design mathematical models required for quantum computing applications

**YOU'LL PRODUCE**

**A Python notebook that verifies unitarity and Hermiticity, computes eigenvalues/eigenvectors, and builds two-qubit states with the tensor product.**

**Tools:** Python 3.13, NumPy, Google Colab / VS Code

**Full step-by-step guide:** Learner Guide — Activity 1: Linear Algebra for Quantum Mechanics in Python


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ACTIVITY 1 · VERIFY & DEBRIEF

# Linear Algebra for Quantum Mechanics in Python

### TEST IT — HOW YOU KNOW IT WORKED

H-dagger H equals the identity, X and Z return real eigenvalues +1 and -1, the Born-rule probability of measuring 0 in |+> is 0.5, and the tensor product produces a 4-component two-qubit state.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC RECAP

# Recap — Quantum Physics Fundamentals

- The five postulates
- Hilbert space
- Unitary operators
- Hermitian operators
- Dirac notation
- Tensor products


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02

TOPIC 02

# Introduction to Quantum Computing Systems

Overview of quantum computers · Commercial systems · Superposition and Entanglement


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MAPS TO LO2, LO4

# Key Concepts — Introduction to Quantum Computing Systems

1

## Qubit vs classical bit

A bit is 0 or 1; a qubit is $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$, allowing superposition of both values.

2

## Bloch sphere

Every pure single-qubit state maps to a point on the unit sphere — the standard geometric picture of qubit state and rotation.

3

## Physical qubits

Superconducting (Josephson junction), trapped-ion and topological qubits trade off speed, coherence time and connectivity.

4

## Commercial platforms

IBM Quantum, Google, IonQ, Rigetti and D-Wave offer cloud access to real quantum processors today.

5

## Superposition

A quantum register of n qubits holds a superposition of all $2^n$ basis states simultaneously — the source of quantum parallelism.

6

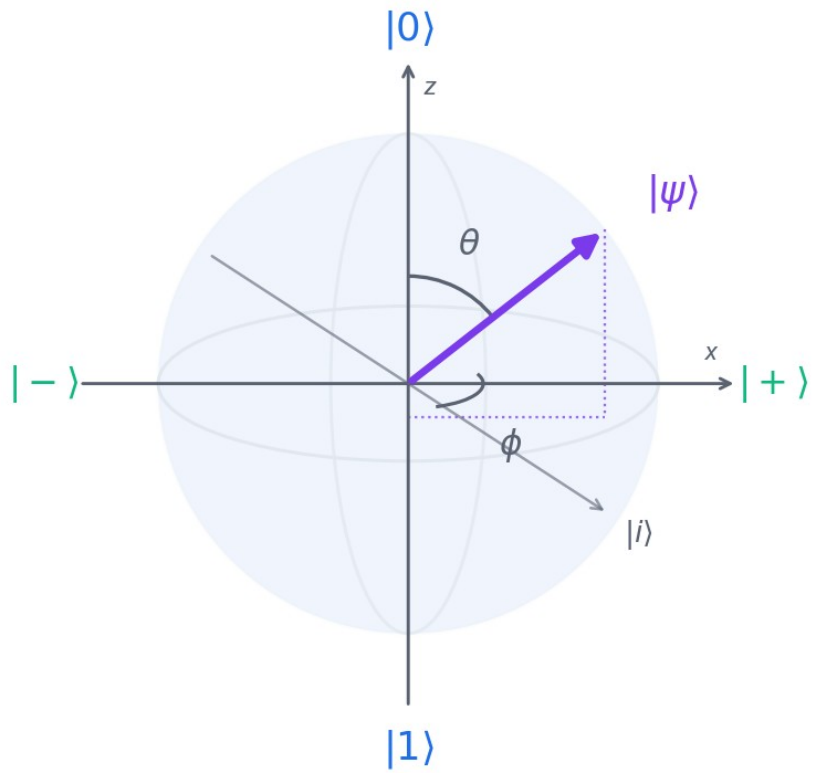
## Entanglement

Entangled states cannot be written as a product of single-qubit states; measuring one qubit instantly correlates with the other.


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GEOMETRY OF A SINGLE QUBIT

# Visualising a Qubit — The Bloch Sphere




$$|\psi\rangle = \cos(\theta/2)\,|0\rangle + e^{i\phi}\sin(\theta/2)\,|1\rangle$$

**North / south poles**

|0⟩ and |1⟩ — the classical computational basis states.

**The equator**

Equal superpositions such as |+⟩ and |−⟩ — maximally uncertain in Z.

**θ sets probability**

$\cos^2(\theta/2)$ is the chance of measuring 0.

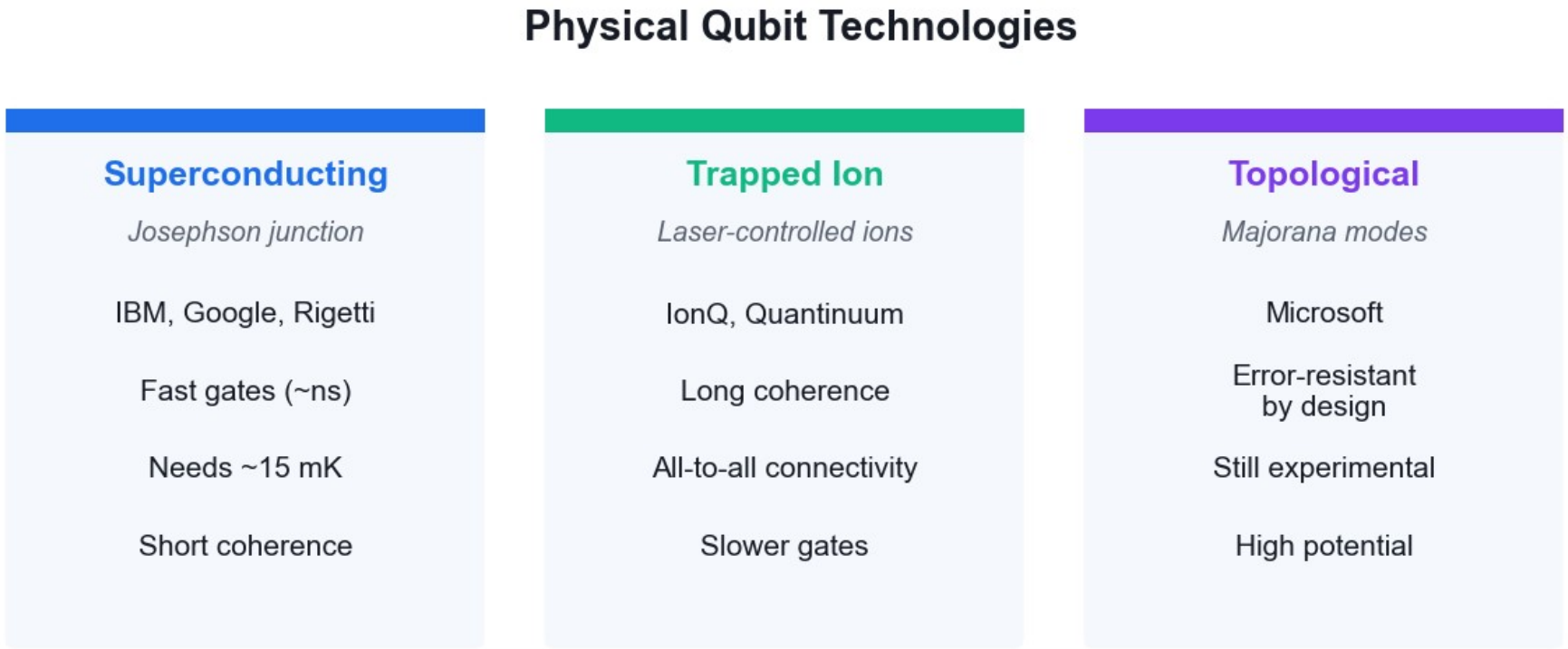
**φ sets phase**

Invisible to a Z measurement, but it drives interference.


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HOW QUBITS ARE ACTUALLY BUILT

# Physical Qubit Technologies

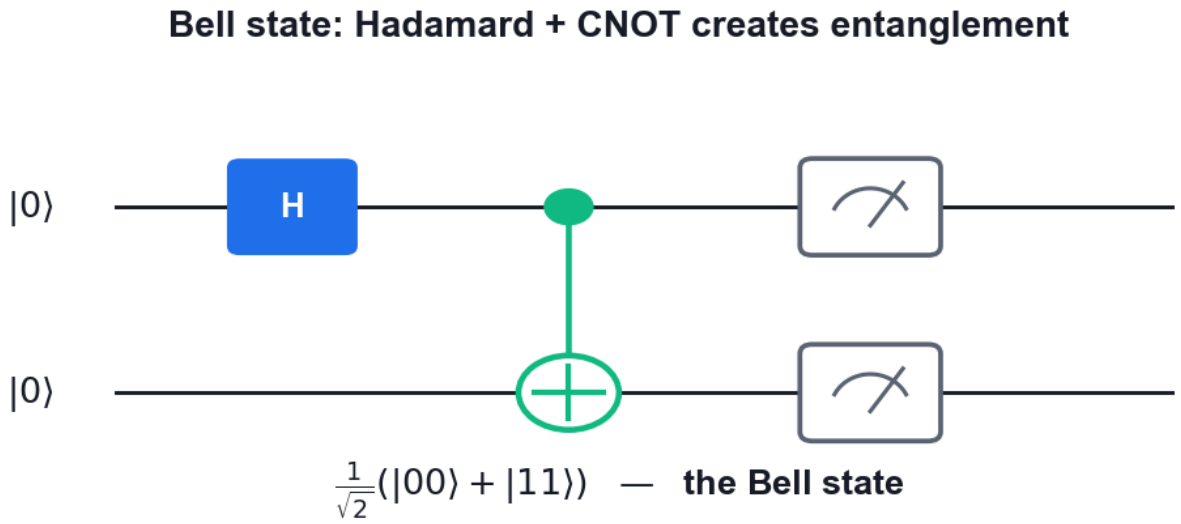



No technology has won yet — each trades gate speed against coherence time and connectivity.


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THE MOST IMPORTANT TWO-QUBIT STATE

# Entanglement — The Bell State




**Two gates**
A Hadamard followed by a CNOT is all it takes.

**Perfect correlation**
Measure 0 on one qubit and the other is certainly 0.

**Not separable**
The state cannot be written as a product of two single-qubit states.

**The resource**
Teleportation, superdense coding and QKD all consume entanglement.


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WHAT YOU'LL DO

# Hands-On Activities — Introduction to Quantum Computing Systems

## Activity 2

- Visualising a Qubit on the Bloch Sphere

## Activity 3

- Superposition and Entanglement on IBM Quantum

## —

- —


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TOPIC 02 · HANDS-ON ACTIVITY

ACTIVITY 2

# Visualising a Qubit on the Bloch Sphere

Build single-qubit states in Qiskit and plot them on the Bloch sphere, so you can see how gates rotate a qubit and what superposition and phase actually look like geometrically.

**MAPS TO** LO2 — Decide on appropriate physical methodologies specific to quantum computing problems

**YOU'LL PRODUCE**

**A set of Bloch-sphere plots showing |0>, |1>, |+>, |-> and an arbitrary Ry-rotated state.**

**Tools:** Qiskit 2.x, matplotlib, Google Colab

**Full step-by-step guide:** Learner Guide — Activity 2: Visualising a Qubit on the Bloch Sphere


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ACTIVITY 2 · VERIFY & DEBRIEF

# Visualising a Qubit on the Bloch Sphere

### TEST IT — HOW YOU KNOW IT WORKED

|0> sits at the north pole, X moves it to the south pole, H places it on the equator at +x, and the S gate rotates it around the equator while leaving the Z-basis probabilities at 50/50.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC 02 · HANDS-ON ACTIVITY

ACTIVITY 3

# Superposition and Entanglement on IBM Quantum

Create your IBM Quantum account, build a superposition circuit and a two-qubit Bell state, and run them on a real quantum processor to see genuine quantum randomness and correlation in the results.

**MAPS TO** LO2, LO4 — Evaluate quantum computing applicability in the business context

**YOU'LL PRODUCE**

**A Bell state circuit executed on real IBM Quantum hardware, with a measured histogram showing the |00> and |11> correlation.**

**Tools:** IBM Quantum Platform, Qiskit Runtime, Python 3.13

**Full step-by-step guide:** Learner Guide — Activity 3: Superposition and Entanglement on IBM Quantum


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ACTIVITY 3 · VERIFY & DEBRIEF

# Superposition and Entanglement on IBM Quantum

### TEST IT — HOW YOU KNOW IT WORKED

The superposition circuit gives roughly 50% |0> and 50% |1>; the Bell circuit gives roughly 50% |00> and 50% |11> with only a small percentage of |01> and |10> counts caused by hardware noise.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC RECAP

# Recap — Introduction to Quantum Computing Systems

- Qubit vs classical bit
- Bloch sphere
- Physical qubits
- Commercial platforms
- Superposition
- Entanglement


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03

TOPIC 03

# Quantum Gates and Circuits

Single Qubit Gates · Multi Qubit Gates · Bell and GHZ States · Phase Kickback


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MAPS TO LO3, LO5

# Key Concepts — Quantum Gates and Circuits

1

## Pauli gates

X (bit flip), Y and Z (phase flip) are self-inverse, Hermitian and unitary — the building blocks of single-qubit logic.

2

## Hadamard gate

H maps $|0\rangle \rightarrow |+\rangle$ and $|1\rangle \rightarrow |-\rangle$, creating the equal superposition that starts nearly every quantum algorithm.

3

## Rotation and phase gates

Rx, Ry, Rz rotate the Bloch vector; S and T are special cases of the phase gate $P(\theta)$.

4

## Multi-qubit gates

CNOT, CZ, controlled-phase, SWAP and Toffoli (CCNOT) create the entanglement that makes quantum computing powerful.

5

## Bell and GHZ states

H followed by CNOT produces a maximally entangled Bell pair; extending to n qubits gives the GHZ state.

6

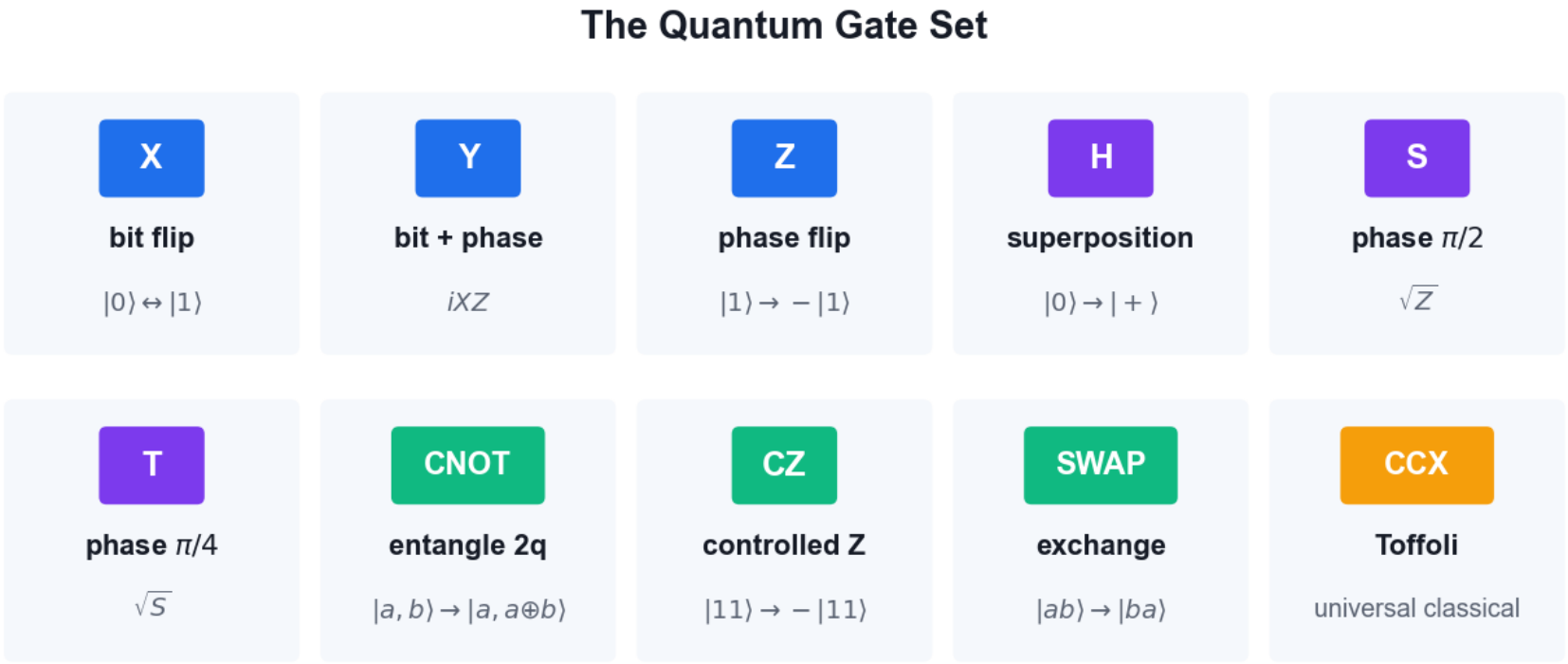
## Phase kickback

A controlled operation on an eigenstate transfers the eigenvalue phase back to the control qubit — the engine of Deutsch, BV and QPE.


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YOUR GATE VOCABULARY

# The Quantum Gate Set




Single-qubit gates rotate the Bloch vector; two-qubit gates create the entanglement that makes quantum computing powerful.


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GATE REFERENCE

# Single vs Multi-Qubit Gates

| Gate | Qubits | Effect | Used for |
|---|---|---|---|
| **X / Y / Z** | 1 | Rotate π about an axis | Bit flip, phase flip |
| **H** | 1 | Basis change | Create superposition |
| **S / T** | 1 | Phase π/2 and π/4 | Fine phase control |
| **Rx / Ry / Rz** | 1 | Arbitrary rotation | Variational ansätze |
| **CNOT** | 2 | Flip target if control is 1 | Entanglement, oracles |
| **CZ / CP** | 2 | Conditional phase | QFT, phase kickback |
| **SWAP** | 2 | Exchange two qubits | Routing on hardware |
| **Toffoli** | 3 | Flip if both controls are 1 | Reversible classical logic |


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THE KEY MECHANISM

# Phase kickback is the trick behind almost every algorithm.

Apply a controlled-U to an eigenstate of U and the eigenvalue phase appears on the CONTROL qubit — that is how Deutsch, Bernstein-Vazirani, phase estimation and Shor all extract their answers.


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WHAT YOU'LL DO

# Hands-On Activities — Quantum Gates and Circuits

## Activity 4

- Building Circuits in IBM Quantum Composer

## Activity 5

- Quantum Gates and Bell/GHZ States with PennyLane

## —

- —


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TOPIC 03 · HANDS-ON ACTIVITY

ACTIVITY 4

# Building Circuits in IBM Quantum Composer

Use the drag-and-drop IBM Quantum Composer to build single- and multi-qubit circuits, watch the state vector and probability panels update live, and export the circuit as OpenQASM.

**MAPS TO** LO5 — Spearhead the application of quantum gates to quantum computing domains

**YOU'LL PRODUCE**

**A set of Composer circuits (X, H, CNOT, Bell state) with their statevector output, exported as OpenQASM 3 code.**

**Tools:** IBM Quantum Composer, OpenQASM 3

**Full step-by-step guide:** Learner Guide — Activity 4: Building Circuits in IBM Quantum Composer


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ACTIVITY 4 · VERIFY & DEBRIEF

# Building Circuits in IBM Quantum Composer

### TEST IT — HOW YOU KNOW IT WORKED

The Composer probability panel shows a 50/50 split between |00> and |11> for the Bell circuit, and the exported OpenQASM contains an h instruction followed by a cx instruction.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC 03 · HANDS-ON ACTIVITY

ACTIVITY 5

# Quantum Gates and Bell/GHZ States with PennyLane

Use PennyLane's quantum-function and QNode model to apply the full single- and multi-qubit gate set, then construct Bell and GHZ entangled states and verify their measurement correlations.

**MAPS TO** LO3, LO5 — Direct two-level quantum models and apply quantum gates

**YOU'LL PRODUCE**

**A PennyLane script that draws each circuit, returns the state vector, and confirms the Bell and GHZ correlations.**

**Tools:** PennyLane, Python 3.13, matplotlib

**Full step-by-step guide:** Learner Guide — Activity 5: Quantum Gates and Bell/GHZ States with PennyLane


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ACTIVITY 5 · VERIFY & DEBRIEF

# Quantum Gates and Bell/GHZ States with PennyLane

### TEST IT — HOW YOU KNOW IT WORKED

The Bell state returns amplitudes of about 0.707 on |00> and |11> and zero elsewhere; the GHZ state returns amplitudes only on |000> and |111>; the phase-kickback circuit shows the eigenvalue phase transferred to the control qubit.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC RECAP

# Recap — Quantum Gates and Circuits

- Pauli gates
- Hadamard gate
- Rotation and phase gates
- Multi-qubit gates
- Bell and GHZ states
- Phase kickback


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04

TOPIC 04

# Quantum Protocols

No-Cloning Theorem · Teleportation · Superdense Coding · BB84 QKD · CHSH Game


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MAPS TO LO3, LO7

# Key Concepts — Quantum Protocols

1

## No-cloning theorem

No unitary can copy an arbitrary unknown quantum state — proved by linearity, and the foundation of quantum key security.

2

## Quantum teleportation

Transfers an unknown qubit state using one shared Bell pair plus two classical bits — no faster-than-light signalling.

3

## Superdense coding

The dual of teleportation: send two classical bits by transmitting a single qubit of a pre-shared entangled pair.

4

## BB84 QKD

Alice and Bob build a shared secret key using random bases; eavesdropping disturbs the states and raises the error rate.

5

## Eavesdropping detection

An intercept-resend attack introduces roughly a 25% error rate in the sifted key, revealing Eve's presence.

6

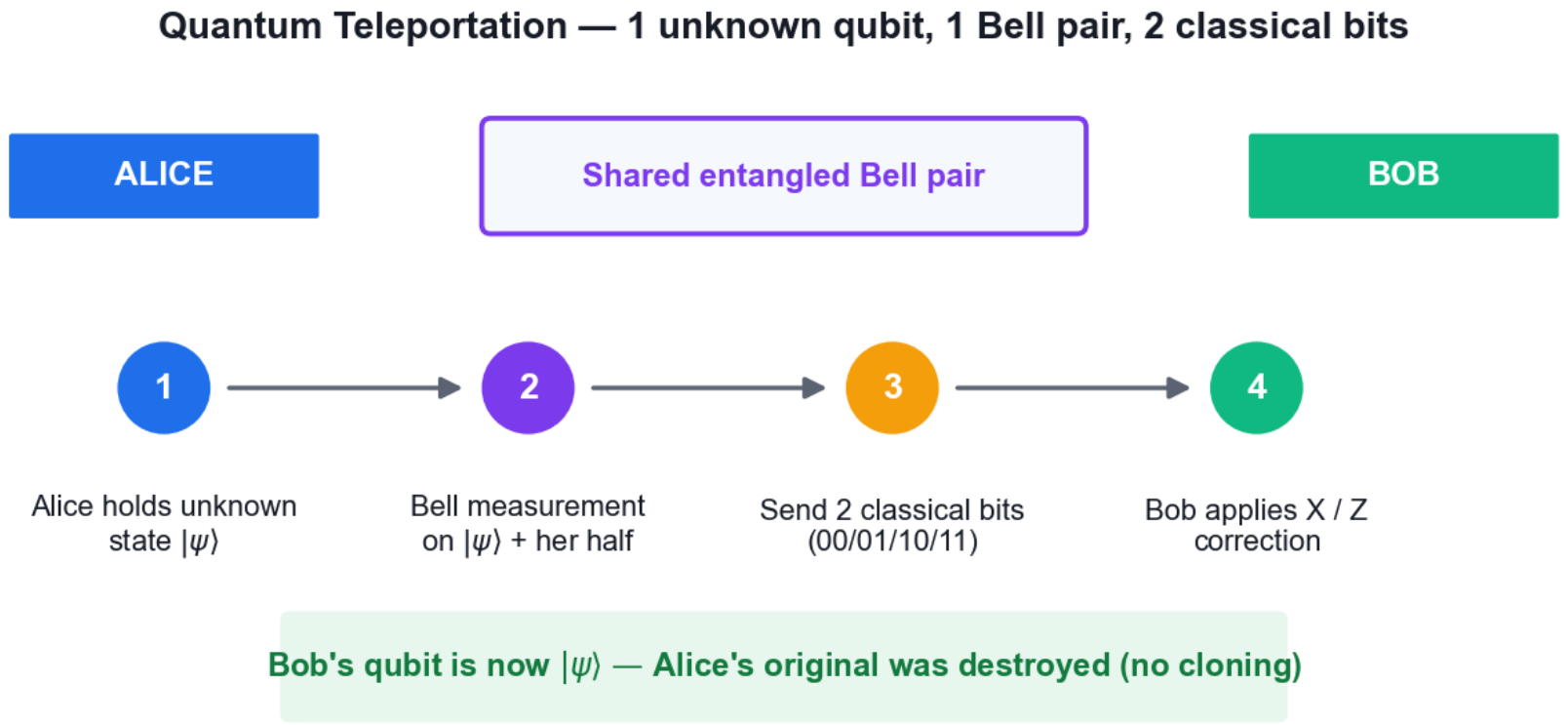
## CHSH game

A quantum strategy using entanglement wins ~85% of the time versus the classical maximum of 75%, disproving local hidden variables.


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MOVING A STATE WITHOUT MOVING A PARTICLE

# Quantum Teleportation




Two classical bits must still travel — so teleportation never beats the speed of light.


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TWO SIDES OF ONE COIN

# Teleportation vs Superdense Coding

## Qubit → classical channel

- Quantum Teleportation
  - Sends: 1 unknown QUBIT state
  - Costs: 1 Bell pair + 2 classical bits
  - Original is destroyed (no cloning)
  - Use: moving quantum data

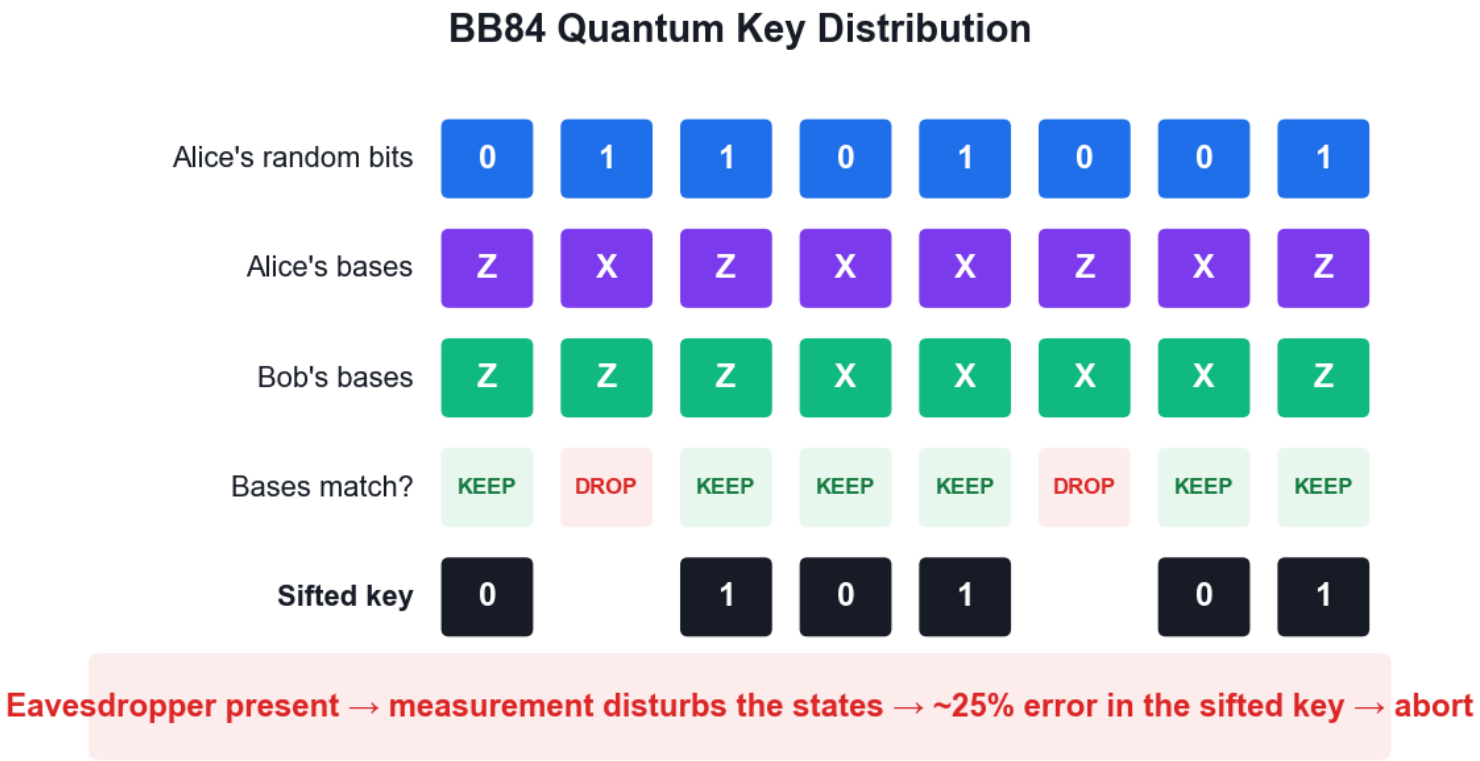
## Classical → qubit channel

- Superdense Coding
  - Sends: 2 CLASSICAL bits
  - Costs: 1 Bell pair + 1 transmitted qubit
  - Encode with I, X, Z or ZX
  - Use: doubling channel capacity


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SECURITY FROM PHYSICS, NOT MATHS

# BB84 Quantum Key Distribution




Security rests on the no-cloning theorem — an eavesdropper cannot copy the qubits without disturbing them.


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WHAT YOU'LL DO

# Hands-On Activities — Quantum Protocols

## Activity 6

- Quantum Teleportation

## Activity 7

- Superdense Coding Protocol

## Activity 8

- BB84 Quantum Key Distribution with Eavesdropping Detection


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TOPIC 04 · HANDS-ON ACTIVITY

ACTIVITY 6

# Quantum Teleportation

Implement the three-qubit teleportation protocol: entangle a shared pair, perform a Bell measurement on the unknown state, and apply the conditional corrections that recreate the original state on Bob's qubit.

**MAPS TO** LO3, LO7 — Synthesize quantum methodologies to address complex problems

**YOU'LL PRODUCE**

**A teleportation circuit run on IBM Quantum hardware that reconstructs an arbitrary input state at the receiving qubit.**

**Tools:** Qiskit 2.x, IBM Quantum Runtime, OpenQASM

**Full step-by-step guide:** Learner Guide — Activity 6: Quantum Teleportation


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ACTIVITY 6 · VERIFY & DEBRIEF

# Quantum Teleportation

### TEST IT — HOW YOU KNOW IT WORKED

The measurement statistics on Bob's qubit q2 match the amplitudes of the state originally prepared on q0, within hardware noise, and the original state no longer exists on q0.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC 04 · HANDS-ON ACTIVITY

ACTIVITY 7

# Superdense Coding Protocol

Implement the dual of teleportation: transmit two classical bits by sending only a single qubit of a pre-shared entangled pair, encoding the message with the I, X, Z and XZ operations.

**MAPS TO** LO3, LO7 — Direct two-level quantum models for quantum computing

**YOU'LL PRODUCE**

**A superdense coding circuit that correctly transmits all four two-bit messages 00, 01, 10 and 11.**

**Tools:** Qiskit 2.x, IBM Quantum Runtime

**Full step-by-step guide:** Learner Guide — Activity 7: Superdense Coding Protocol


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ACTIVITY 7 · VERIFY & DEBRIEF

# Superdense Coding Protocol

### TEST IT — HOW YOU KNOW IT WORKED

Each of the four encodings decodes to its intended message — identity gives 00, X gives 01, Z gives 10 and ZX gives 11 — demonstrating two classical bits carried by one transmitted qubit.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC 04 · HANDS-ON ACTIVITY

ACTIVITY 8

# BB84 Quantum Key Distribution with Eavesdropping Detection

Run the BB84 protocol end to end: random bit and basis selection, quantum transmission, basis sifting and key extraction — then repeat it with an eavesdropper present and measure the error rate that exposes the attack.

**MAPS TO** LO7 — Synthesize quantum methodologies to address complex problems

**YOU'LL PRODUCE**

**A working BB84 implementation producing a shared secret key, plus a comparison run showing the ~25% error rate introduced by an intercept-resend eavesdropper.**

**Tools:** Qiskit 2.x, IBM Quantum Runtime, Python

**Full step-by-step guide:** Learner Guide — Activity 8: BB84 Quantum Key Distribution with Eavesdropping Detection


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ACTIVITY 8 · VERIFY & DEBRIEF

# BB84 Quantum Key Distribution with Eavesdropping Detection

### TEST IT — HOW YOU KNOW IT WORKED

Without an eavesdropper the sifted key error rate is near zero (bounded only by hardware noise); with the intercept-resend eavesdropper the error rate rises to roughly 25%, which is the signal to abort and discard the key.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC RECAP

# Recap — Quantum Protocols

- No-cloning theorem
- Quantum teleportation
- Superdense coding
- BB84 QKD
- Eavesdropping detection
- CHSH game


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05

TOPIC 05

# Quantum Algorithms

Deutsch · Deutsch-Jozsa · Bernstein-Vazirani · Grover · Quantum Fourier Transform


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MAPS TO LO6, LO7

# Key Concepts — Quantum Algorithms

1

## The query (oracle) model

Algorithms are compared by how many times they query a black-box function — the standard measure of quantum advantage.

2

## Deutsch algorithm

Decides whether $f:\{0,1\}\to\{0,1\}$ is constant or balanced with ONE query; classically two are required.

3

## Deutsch-Jozsa

Extends the idea to n bits — one query versus $2^{n-1}+1$ classical queries in the worst case: exponential separation.

4

## Bernstein-Vazirani

Recovers an n-bit hidden string s from $f(x) = s\cdot x \pmod 2$ in a single query instead of n.

5

## Grover search

Amplitude amplification finds a marked item in an unstructured database in $O(\sqrt{N})$ instead of $O(N)$ — a quadratic speedup.

6

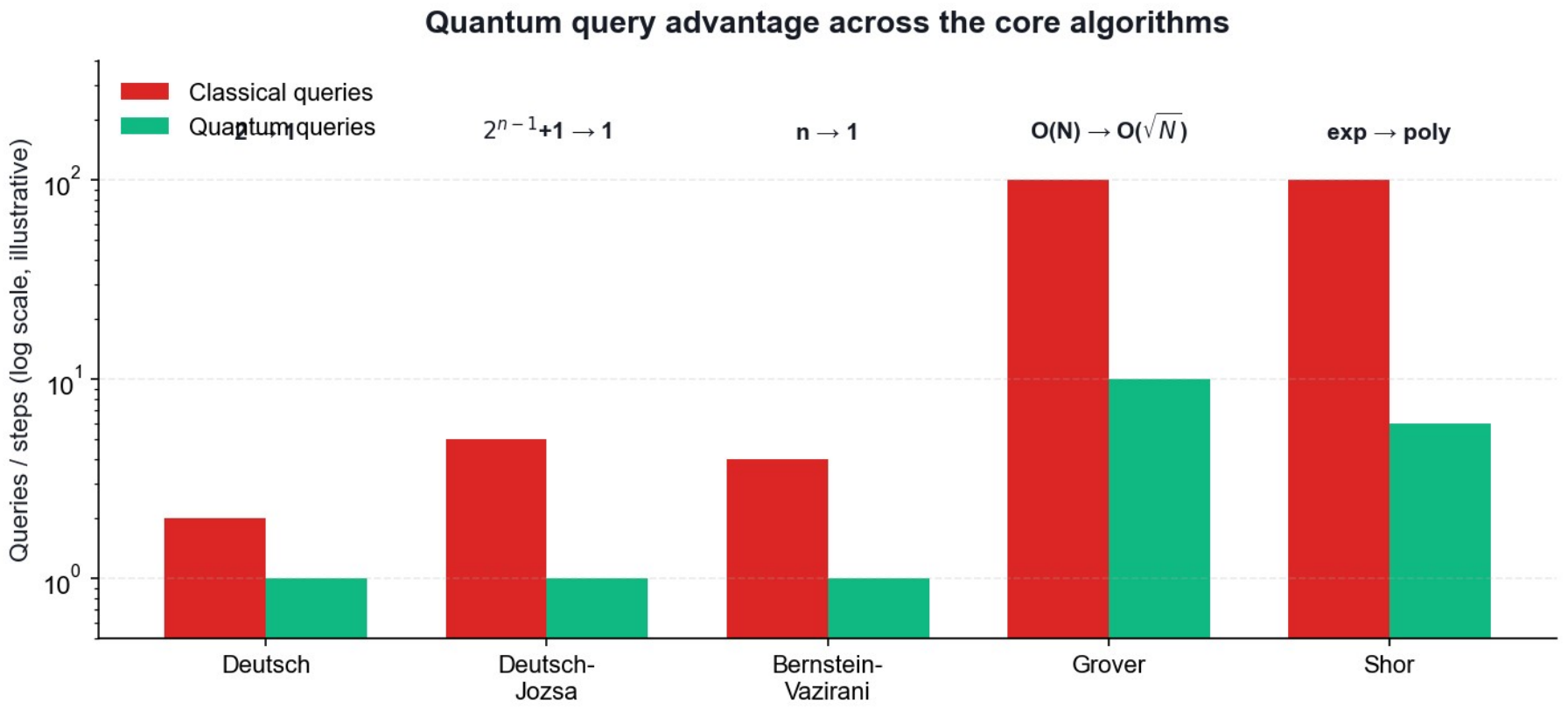
## Quantum Fourier Transform

The quantum analogue of the DFT runs in $O(n^2)$ gates and underpins phase estimation and Shor's algorithm.


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WHY THESE ALGORITHMS MATTER

# Quantum Query Advantage

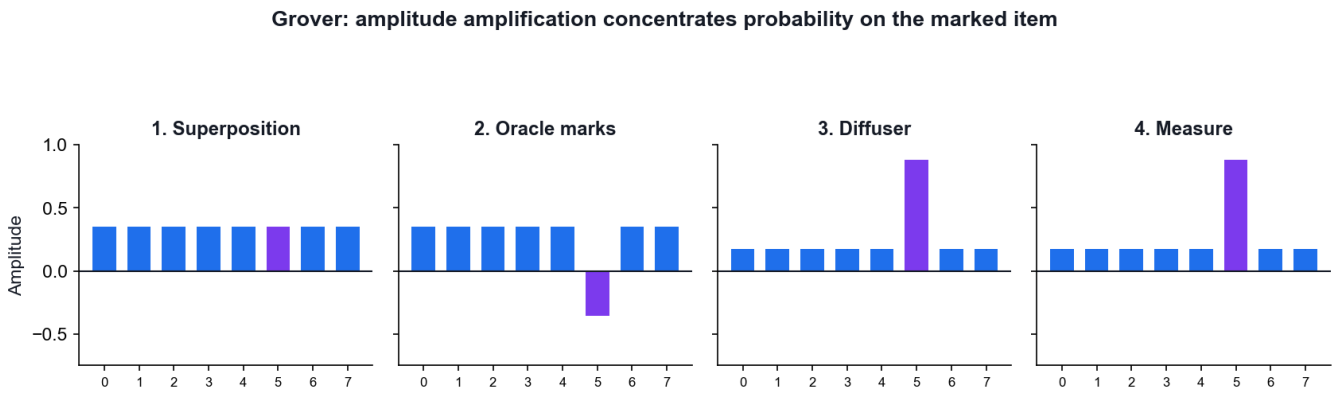



Query counts are illustrative — the point is the SCALING, not the absolute numbers.


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HOW GROVER ACTUALLY WORKS

# Grover — Amplitude Amplification




**Start**
All N states share an equal amplitude of $1/\sqrt{N}$.

**Oracle**
Flips the SIGN of the marked state — invisible to measurement so far.

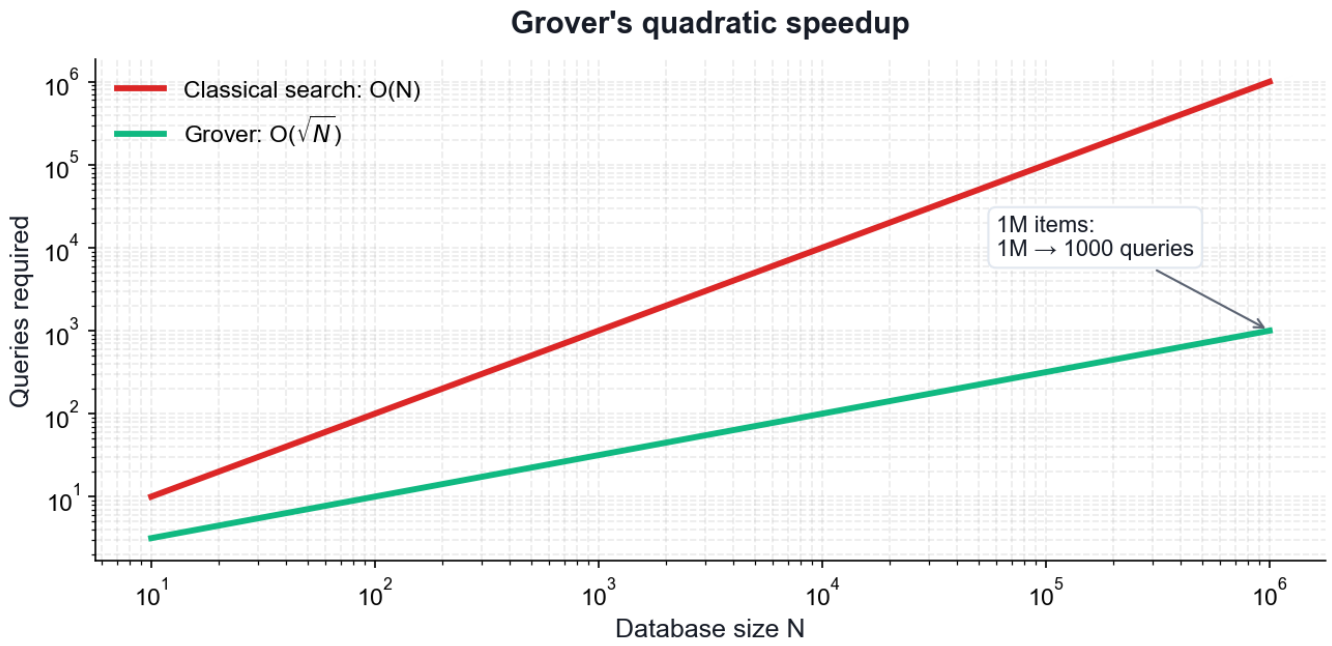
**Diffuser**
Reflects every amplitude about the mean, growing the marked one.

**Repeat**
About $(\pi/4)\sqrt{N}$ times — then measure.


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THE SCALING

# Grover's Quadratic Speedup




**Classical**

Checking an unstructured list takes O(N) queries.

**Grover**

Finds the marked item in O(√N) queries.

**1M items**

1,000,000 checks become roughly 1,000.

**Not exponential**

A real speedup, but far smaller than Shor's.


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WHAT YOU'LL DO

# Hands-On Activities — Quantum Algorithms

## Activities 9–10

- Deutsch and Deutsch-Jozsa Algorithms
- Bernstein-Vazirani Algorithm

## Activities 11–12

- Grover's Search Algorithm
- Quantum Fourier Transform

## —

- —


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TOPIC 05 · HANDS-ON ACTIVITY

ACTIVITY 9

# Deutsch and Deutsch-Jozsa Algorithms

Implement the first algorithms to show a provable quantum advantage: decide whether a black-box function is constant or balanced with a single query, then generalise from one input bit to n.

**MAPS TO** LO6 — Establish guidelines for selecting quantum computing algorithms

**YOU'LL PRODUCE**

**Deutsch and Deutsch-Jozsa circuits with all four oracle types, run on IBM Quantum hardware, correctly classifying each function in one query.**

**Tools:** Qiskit 2.x, IBM Quantum Runtime, OpenQASM

**Full step-by-step guide:** Learner Guide — Activity 9: Deutsch and Deutsch-Jozsa Algorithms


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ACTIVITY 9 · VERIFY & DEBRIEF

# Deutsch and Deutsch-Jozsa Algorithms

### TEST IT — HOW YOU KNOW IT WORKED

The Deutsch circuit measures 0 for both constant oracles and 1 for both balanced oracles; Deutsch-Jozsa returns the all-zeros string for a constant function and a non-zero string for a balanced function — each using exactly one oracle query.

### DISCUSS WITH YOUR TRAINER

**What did the result show?**

Restate the outcome in your own words.

**Why does it work?**

Name the quantum principle the activity relies on.

**Where would you use it?**

Give one realistic application in your own field.


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TOPIC 05 · HANDS-ON ACTIVITY

ACTIVITY 10

# Bernstein-Vazirani Algorithm

Recover an n-bit hidden string in a single oracle query. Classically you would need n queries — one per bit — making this a clean demonstration of linear-to-constant query speedup.

**MAPS TO** LO6 — Establish guidelines for selecting quantum computing algorithms

**YOU'LL PRODUCE**

**A Bernstein-Vazirani circuit that recovers a chosen hidden string such as 1011 in one query on real hardware.**

**Tools:** Qiskit 2.x, IBM Quantum Runtime

**Full step-by-step guide:** Learner Guide — Activity 10: Bernstein-Vazirani Algorithm


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ACTIVITY 10 · VERIFY & DEBRIEF

# Bernstein-Vazirani Algorithm

**TEST IT — HOW YOU KNOW IT WORKED**

The most frequent measured bit string equals the hidden string s that was encoded in the oracle, recovered with a single oracle query regardless of the length of s.

**DISCUSS WITH YOUR TRAINER**

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC 05 · HANDS-ON ACTIVITY

ACTIVITY 11

# Grover's Search Algorithm

Implement amplitude amplification to find a marked item in an unstructured search space, using the phase oracle and the diffuser, and see the quadratic O(sqrt(N)) speedup for 2-qubit and 3-qubit searches.

**MAPS TO** LO6, LO7 — Select algorithms and synthesize quantum methodologies

**YOU'LL PRODUCE**

**Grover circuits for 2 and 3 qubits that amplify the marked state to a high measurement probability on real hardware.**

**Tools:** Qiskit 2.x, IBM Quantum Runtime, OpenQASM

**Full step-by-step guide:** Learner Guide — Activity 11: Grover's Search Algorithm


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ACTIVITY 11 · VERIFY & DEBRIEF

# Grover's Search Algorithm

### TEST IT — HOW YOU KNOW IT WORKED

For 2 qubits the marked state is measured with near-certainty after one Grover iteration; for 3 qubits it dominates the histogram after two iterations; applying too many iterations visibly reduces the success probability.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC 05 · HANDS-ON ACTIVITY

ACTIVITY 12

# Quantum Fourier Transform

Build the QFT circuit from Hadamard and controlled-phase gates, verify it against the classical DFT, and understand why it is the engine behind phase estimation and Shor's algorithm.

**MAPS TO** LO6, LO7 — Establish guidelines for selecting quantum computing algorithms

**YOU'LL PRODUCE**

**QFT circuits for 3, 4 and 5 qubits, validated against a classical DFT reference and run on IBM Quantum hardware.**

**Tools:** Qiskit 2.x, NumPy, IBM Quantum Runtime, OpenQASM

**Full step-by-step guide:** Learner Guide — Activity 12: Quantum Fourier Transform


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ACTIVITY 12 · VERIFY & DEBRIEF

# Quantum Fourier Transform

### TEST IT — HOW YOU KNOW IT WORKED

Your hand-built QFT circuit produces the same state vector as Qiskit's built-in QFT and matches a normalised classical FFT of the same input, using on the order of n^2 gates.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC RECAP

# Recap — Quantum Algorithms

- The query (oracle) model
- Deutsch algorithm
- Deutsch-Jozsa
- Bernstein-Vazirani
- Grover search
- Quantum Fourier Transform


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06

TOPIC 06

# Quantum Cryptography and Quantum Safety

Classical Cryptography · The Quantum Threat · Shor Algorithm · Post-Quantum Cryptography


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MAPS TO LO4, LO7

# Key Concepts — Quantum Cryptography and Quantum Safety

1

## Goals of cryptography

Confidentiality, integrity, authentication and non-repudiation are the four properties every scheme must address.

2

## Classical schemes

Symmetric (AES), asymmetric (RSA, ECC) and hash functions (SHA-256) rest on problems that are hard for classical computers.

3

## The quantum threat

Shor's algorithm breaks RSA and ECC in polynomial time; Grover halves the effective key length of symmetric ciphers.

4

## Shor's algorithm

Reduces factoring to period-finding, then uses the QFT to extract the period — turning a hard problem into an easy one.

5

## Harvest now, decrypt later

Adversaries can record encrypted traffic today and decrypt it once a cryptographically relevant quantum computer exists.

6

## Post-quantum cryptography

NIST has standardised ML-KEM (Kyber), ML-DSA (Dilithium), SLH-DSA (SPHINCS+) and HQC as quantum-resistant replacements.


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WHAT BREAKS AND WHAT SURVIVES

# The Quantum Threat to Cryptography

## The Quantum Threat to Today's Cryptography

| BROKEN by Shor | WEAKENED by Grover | QUANTUM-SAFE (NIST) |
|---|---|---|
| RSA-2048 | AES-128 → 64-bit | ML-KEM (FIPS 203) |
| ECC / ECDSA | SHA-256 → 128-bit | ML-DSA (FIPS 204) |
| Diffie-Hellman | AES-256 → safe | SLH-DSA (FIPS 205) |
| DSA | SHA-512 → safe | HQC (2025) |
| Polynomial time — public-key crypto fails | Effective key length is halved — double it | Standardised replacements |

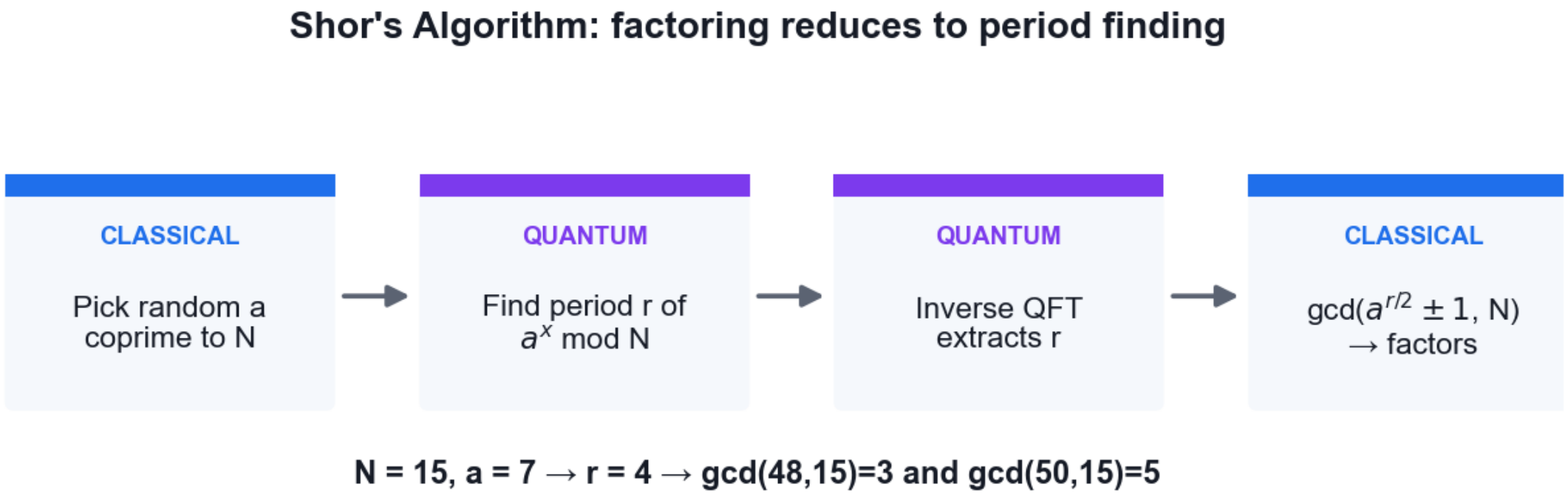
**Harvest now, decrypt later — data captured today is at risk the moment a CRQC arrives**

Symmetric crypto survives by doubling key length; public-key crypto must be replaced outright.


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FACTORING VIA PERIOD FINDING

# Shor's Algorithm




Only the period-finding step is quantum — the rest is classical number theory.


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# Classical Cryptography — What You Must Know

| Type | Example | Key | Broken by |
|---|---|---|---|
| **Symmetric** | AES-256 | One shared secret key | Grover (halves strength) |
| **Asymmetric** | RSA-2048, ECC | Public + private key pair | Shor (fully broken) |
| **Hash** | SHA-256 | No key | Grover (halves strength) |
| **PQC KEM** | ML-KEM (Kyber) | Lattice-based | Not known to be broken |
| **PQC signature** | ML-DSA, SLH-DSA | Lattice / hash-based | Not known to be broken |


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WHY MIGRATE BEFORE THE THREAT ARRIVES

# Harvest now, decrypt later.

An adversary can record your encrypted traffic today and decrypt it the day a cryptographically relevant quantum computer exists. Any secret that must stay confidential for a decade is already at risk.


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WHAT YOU'LL DO

# Hands-On Activities — Quantum Cryptography and Quantum Safety

## Activity 13

- Shor's Algorithm and the Quantum Threat to RSA

## Activity 14

- Post-Quantum Cryptography Migration Assessment

## —

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TOPIC 06 · HANDS-ON ACTIVITY

ACTIVITY 13

# Shor's Algorithm and the Quantum Threat to RSA

See how factoring reduces to period finding, run the quantum period-finding circuit that sits at the heart of Shor's algorithm, and recover the factors of N = 15 from the measured period.

**MAPS TO** LO7 — Synthesize quantum methodologies to address complex problems

**YOU'LL PRODUCE**

**A period-finding circuit whose measured output yields the period r, from which the factors of N = 15 are classically recovered.**

**Tools:** Qiskit 2.x, IBM Quantum Runtime, Python, OpenQASM

**Full step-by-step guide:** Learner Guide — Activity 13: Shor's Algorithm and the Quantum Threat to RSA


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ACTIVITY 13 · VERIFY & DEBRIEF

# Shor's Algorithm and the Quantum Threat to RSA

### TEST IT — HOW YOU KNOW IT WORKED

The measured phase yields period r = 4 for a = 7 and N = 15, and the classical post-processing step recovers the correct factors 3 and 5.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC 06 · HANDS-ON ACTIVITY

ACTIVITY 14

# Post-Quantum Cryptography Migration Assessment

Assess which cryptography in a realistic system is broken by a quantum computer, map each algorithm to its NIST post-quantum replacement, and produce a prioritised migration plan against the harvest-now-decrypt-later risk.

**MAPS TO** LO4, LO7 — Evaluate quantum applicability in the business context

**YOU'LL PRODUCE**

**A completed cryptographic inventory and a prioritised PQC migration plan for a sample organisation.**

**Tools:** NIST FIPS 203/204/205, OpenSSL, Python

**Full step-by-step guide:** Learner Guide — Activity 14: Post-Quantum Cryptography Migration Assessment


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ACTIVITY 14 · VERIFY & DEBRIEF

# Post-Quantum Cryptography Migration Assessment

### TEST IT — HOW YOU KNOW IT WORKED

Your inventory correctly separates Shor-broken asymmetric algorithms from Grover-weakened symmetric ones, maps each to the correct NIST FIPS standard, and ranks migration priority by how long the data must remain confidential.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC RECAP

# Recap — Quantum Cryptography and Quantum Safety

- Goals of cryptography
- Classical schemes
- The quantum threat
- Shor's algorithm
- Harvest now, decrypt later
- Post-quantum cryptography


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07

TOPIC 07

# Quantum Computing Applications and Use Cases

Finance · Healthcare · Science


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MAPS TO LO4

# Key Concepts — Quantum Computing Applications and Use Cases

1

## Finance

Portfolio optimisation, derivative pricing via amplitude estimation, credit risk and fraud detection are the leading finance use cases.

2

## Healthcare and life sciences

Molecular simulation with VQE, protein folding and accelerated drug discovery target problems classical HPC cannot reach.

3

## Science and materials

Simulating catalysts, superconductors, nitrogen fixation and battery chemistry is quantum computing's most natural application.

4

## Optimisation and logistics

QAOA addresses routing, scheduling and Max-Cut style combinatorial problems across supply chains and energy grids.

5

## Quantum machine learning

Quantum kernels and variational classifiers explore feature spaces that are expensive to compute classically.

6

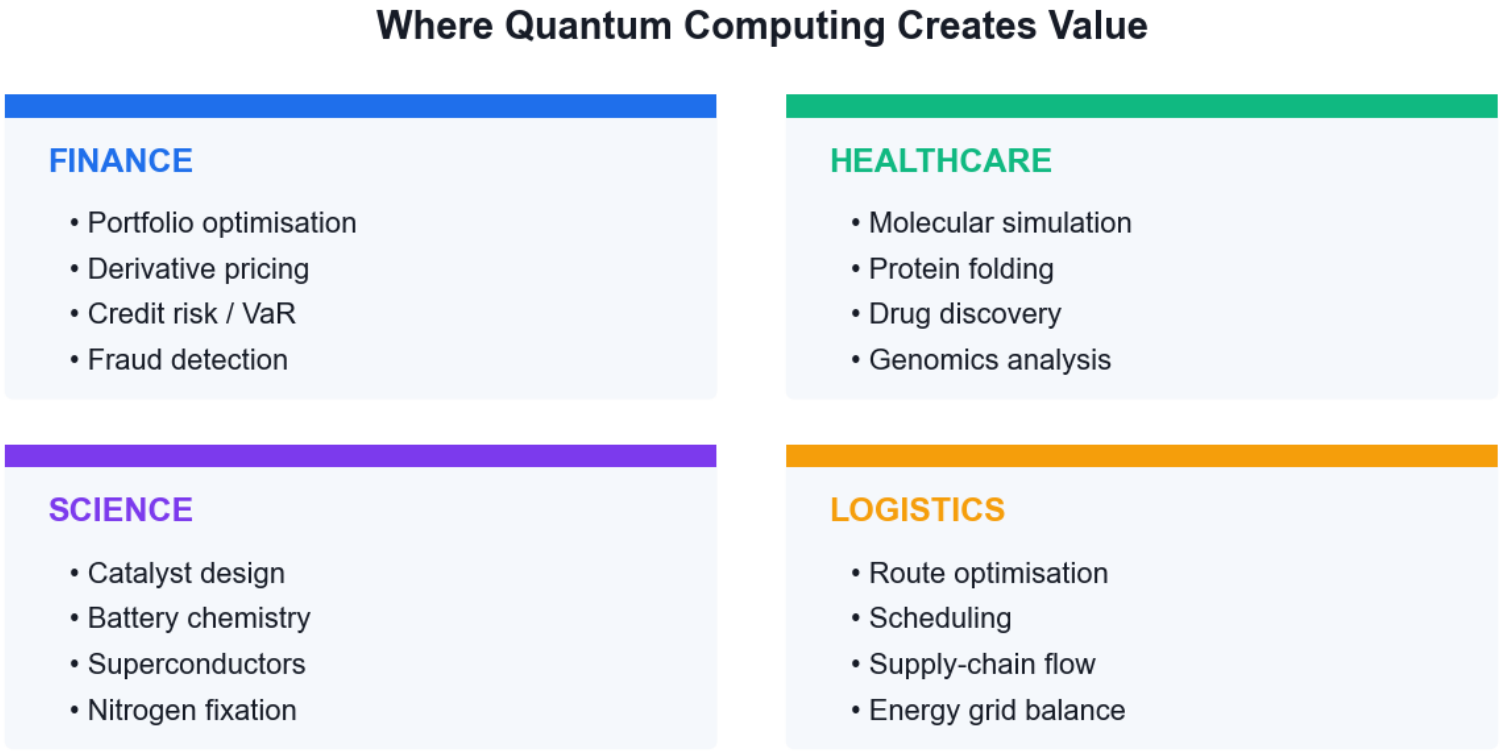
## Assessing business value

Evaluate a use case on problem structure, data size, required qubit count and whether a classical method already suffices.


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USE CASES BY SECTOR

# Where Quantum Creates Value

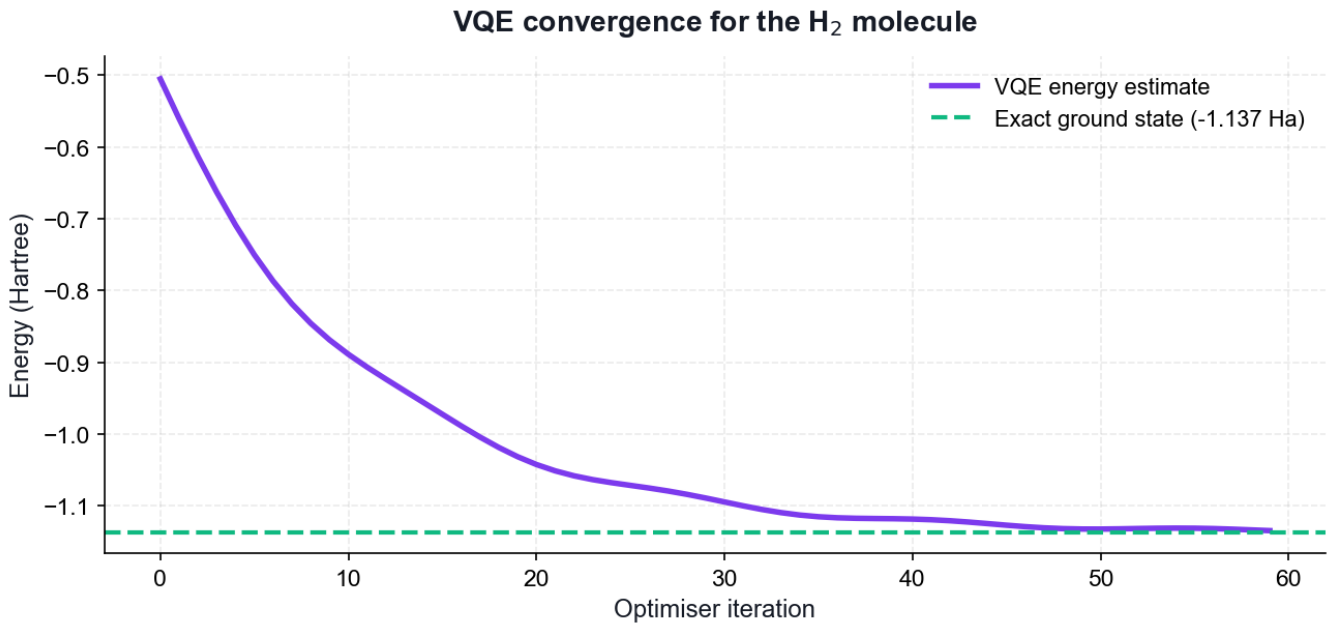



The best candidates are problems with quantum structure — chemistry and optimisation — not big-data problems.


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THE FLAGSHIP NEAR-TERM APPLICATION

# VQE — Simulating Molecules




### Hybrid loop

A quantum circuit prepares the trial state; a classical optimiser tunes it.

### Noise tolerant

Shallow circuits make VQE practical on today's NISQ hardware.

### The target

Ground-state energy — the quantity that drives chemistry.

### The prize

Catalysts, batteries and drug candidates simulated exactly.


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A PRACTICAL CHECKLIST FOR LO4

# Assessing a Quantum Use Case

1

### Does the problem have quantum structure?

Chemistry and optimisation map naturally; generic data processing does not.

2

### How big is the data?

Loading large classical datasets into a quantum computer is itself a bottleneck.

3

### What qubit count is needed?

Estimate logical qubits, then multiply for error-correction overhead.

4

### Is there a good classical method?

If a classical heuristic already solves it well, quantum adds nothing.

5

### What is the time horizon?

NISQ demos today; fault-tolerant advantage is a 2030s proposition.

6

### What is the cost of waiting?

For cryptography the answer is high — migrate now.


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WHAT YOU'LL DO

# Hands-On Activities — Quantum Computing Applications and Use Cases

## Activity 15

- Molecular Simulation with VQE (Healthcare & Science)

## Activity 16

- Portfolio Optimisation with QAOA (Finance)

## —

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ACTIVITY 15

# Molecular Simulation with VQE (Healthcare & Science)

Run the Variational Quantum Eigensolver to compute the ground-state energy of the hydrogen molecule — the flagship near-term use case for drug discovery, catalysis and materials science.

**MAPS TO** LO4, LO7 — Evaluate quantum computing applicability in the business context

**YOU'LL PRODUCE**

**A VQE run that converges to the H2 ground-state energy, with an energy-versus-iteration convergence plot.**

**Tools:** Qiskit 2.x, SciPy optimiser, matplotlib, IBM Quantum

**Full step-by-step guide:** Learner Guide — Activity 15: Molecular Simulation with VQE (Healthcare & Science)


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ACTIVITY 15 · VERIFY & DEBRIEF

# Molecular Simulation with VQE (Healthcare & Science)

### TEST IT — HOW YOU KNOW IT WORKED

The VQE optimisation converges to an energy close to the exact H2 ground-state value of about -1.137 Hartree, and the convergence plot shows the energy decreasing monotonically toward that limit.

### DISCUSS WITH YOUR TRAINER

**What did the result show?**

Restate the outcome in your own words.

**Why does it work?**

Name the quantum principle the activity relies on.

**Where would you use it?**

Give one realistic application in your own field.


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TOPIC 07 · HANDS-ON ACTIVITY

ACTIVITY 16

# Portfolio Optimisation with QAOA (Finance)

Apply the Quantum Approximate Optimization Algorithm to a Max-Cut style combinatorial problem, the same structure that underlies portfolio selection, routing and scheduling in financial services.

**MAPS TO** LO4, LO6 — Evaluate applicability and select appropriate algorithms

**YOU'LL PRODUCE**

**A QAOA run that finds the optimal cut on a small graph, with the solution quality compared against brute force.**

**Tools:** Qiskit 2.x, NetworkX, SciPy, matplotlib

**Full step-by-step guide:** Learner Guide — Activity 16: Portfolio Optimisation with QAOA (Finance)


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ACTIVITY 16 · VERIFY & DEBRIEF

# Portfolio Optimisation with QAOA (Finance)

### TEST IT — HOW YOU KNOW IT WORKED

The most frequently sampled bit string from the optimised QAOA circuit corresponds to the true maximum cut found by brute force, and the approximation ratio improves as the number of layers p increases.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC RECAP

# Recap — Quantum Computing Applications and Use Cases

- Finance
- Healthcare and life sciences
- Science and materials
- Optimisation and logistics
- Quantum machine learning
- Assessing business value


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08

TOPIC 08

# Quantum Error Correction

Bit and Phase Flip Errors · QEC Algorithms · Fault Tolerance · The Future


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MAPS TO LO7

# Key Concepts — Quantum Error Correction

1

## Why QEC is needed

Qubits decohere and gates are imperfect; without correction, errors accumulate long before a useful algorithm completes.

2

## Quantum-specific challenges

No-cloning forbids simple copying, errors are continuous rather than discrete, and measurement destroys the state.

3

## Syndrome measurement

Ancilla qubits are measured to reveal WHICH error occurred without ever measuring — and collapsing — the logical state.

4

## Bit-flip and phase-flip codes

Three-qubit repetition codes correct X errors; conjugating with Hadamards turns phase flips into bit flips.

5

## Shor, Steane and surface codes

Shor's 9-qubit code corrects arbitrary single-qubit errors; Steane uses 7; surface codes are today's leading practical approach.

6

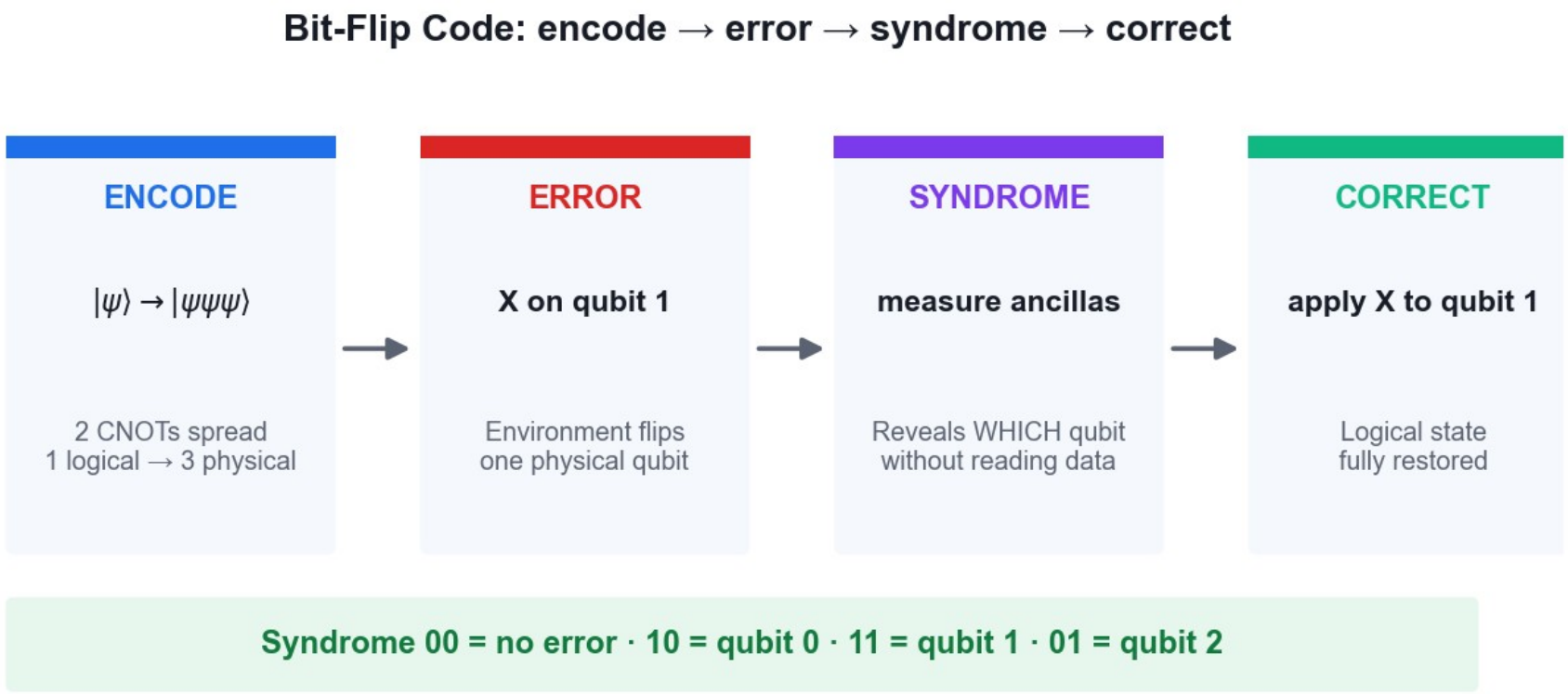
## Fault tolerance

Below the error threshold, concatenating codes suppresses errors faster than they accumulate — the path to useful quantum computers.


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ENCODE · ERROR · SYNDROME · CORRECT

# Bit-Flip Code — The QEC Cycle

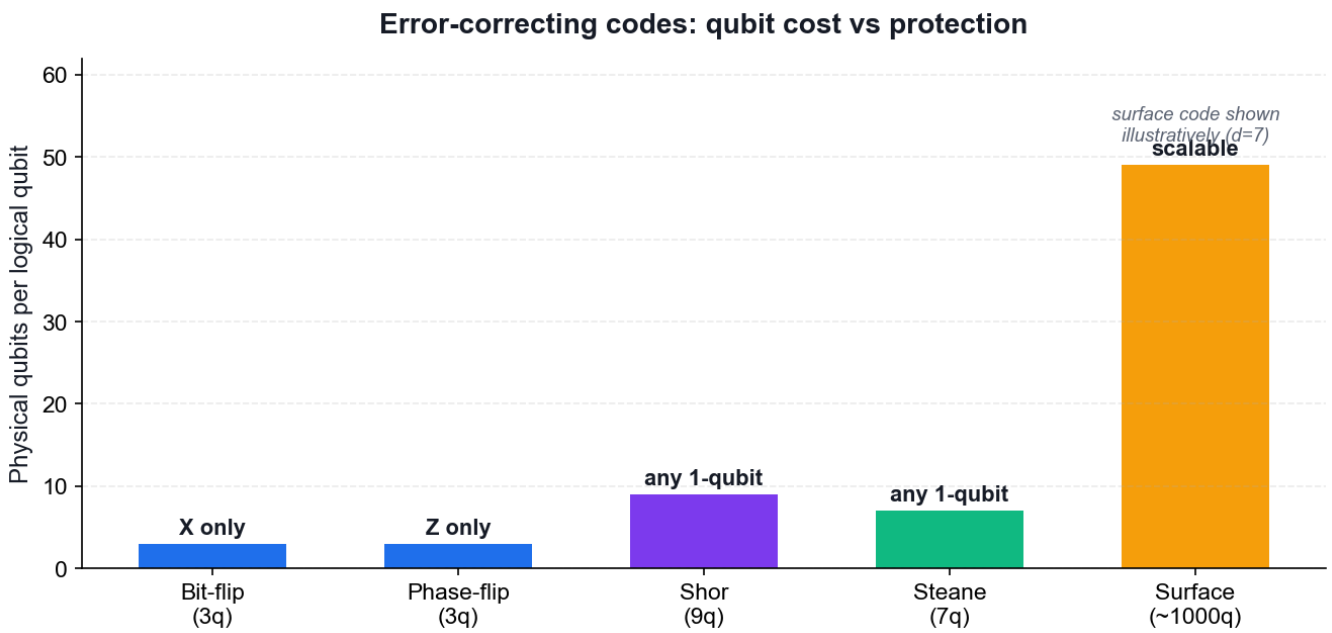



The syndrome reveals WHICH qubit flipped without ever measuring — and destroying — the logical state.


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QUBIT COST VS PROTECTION

# Error-Correcting Codes Compared




### 3-qubit codes

Correct either bit flips or phase flips, but not both.

### Shor 9-qubit

The first code to correct an arbitrary single-qubit error.

### Steane 7-qubit

A CSS code achieving the same protection more efficiently.

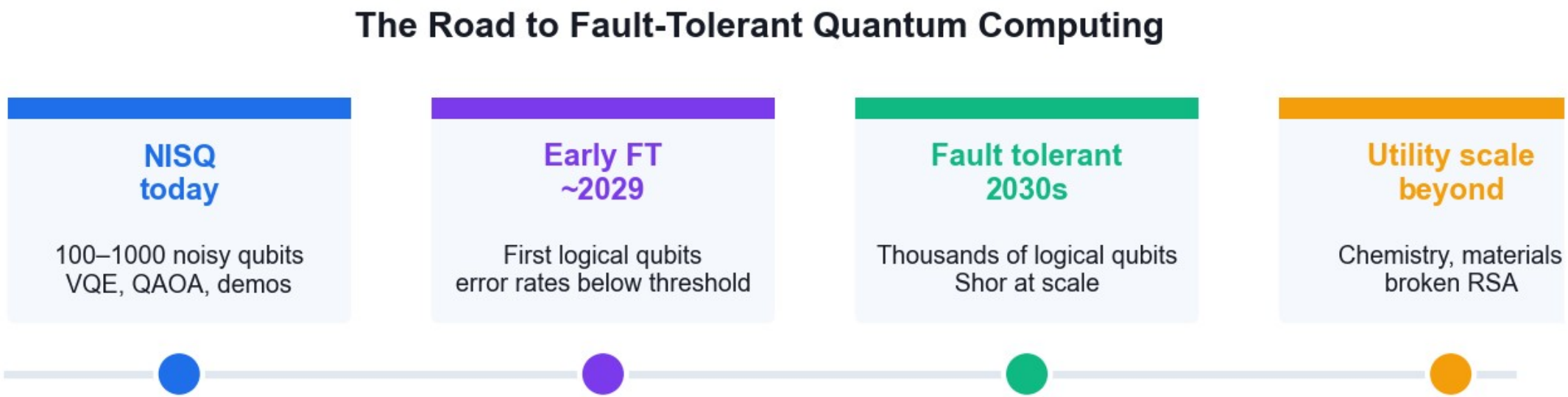
### Surface codes

2D nearest-neighbour layout — today's leading practical approach.


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WHERE THE INDUSTRY IS HEADED

# The Road to Fault Tolerance




Below the error threshold, adding more physical qubits suppresses errors faster than they accumulate.


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WHAT YOU'LL DO

# Hands-On Activities — Quantum Error Correction

## Activity 17

- Bit-Flip and Phase-Flip Error Correction Codes

## Activity 18

- Shor 9-Qubit and Steane 7-Qubit Codes

## —

- —


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TOPIC 08 · HANDS-ON ACTIVITY

ACTIVITY 17

# Bit-Flip and Phase-Flip Error Correction Codes

Build the three-qubit repetition codes that detect and correct X and Z errors: encode a logical qubit, inject an error, measure the syndrome without collapsing the state, and apply the correction.

**MAPS TO** LO7 — Synthesize quantum methodologies to address complex problems

**YOU'LL PRODUCE**

**Working bit-flip and phase-flip code circuits that correctly identify and repair an injected single-qubit error.**

**Tools:** Qiskit 2.x, IBM Quantum Runtime, OpenQASM

**Full step-by-step guide:** Learner Guide — Activity 17: Bit-Flip and Phase-Flip Error Correction Codes


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ACTIVITY 17 · VERIFY & DEBRIEF

# Bit-Flip and Phase-Flip Error Correction Codes

### TEST IT — HOW YOU KNOW IT WORKED

The syndrome measurement correctly identifies which qubit carried the injected error without collapsing the logical state, and applying the indicated correction restores the original logical qubit.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC 08 · HANDS-ON ACTIVITY

ACTIVITY 18

# Shor 9-Qubit and Steane 7-Qubit Codes

Step up to codes that correct arbitrary single-qubit errors: Shor's 9-qubit concatenated code and Steane's more efficient 7-qubit CSS code, then compare their qubit overhead and discuss the road to fault tolerance.

**MAPS TO** LO7 — Synthesize quantum methodologies to address complex problems

**YOU'LL PRODUCE**

**Working Shor 9-qubit and Steane 7-qubit code circuits, with a comparison of qubit cost against protection offered.**

**Tools:** Qiskit 2.x, IBM Quantum Runtime, OpenQASM

**Full step-by-step guide:** Learner Guide — Activity 18: Shor 9-Qubit and Steane 7-Qubit Codes


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ACTIVITY 18 · VERIFY & DEBRIEF

# Shor 9-Qubit and Steane 7-Qubit Codes

### TEST IT — HOW YOU KNOW IT WORKED

Both the Shor 9-qubit and Steane 7-qubit circuits detect and correct an arbitrary injected single-qubit error (X, Y or Z), and you can state the qubit overhead each code requires.

## DISCUSS WITH YOUR TRAINER

### What did the result show?

Restate the outcome in your own words.

### Why does it work?

Name the quantum principle the activity relies on.

### Where would you use it?

Give one realistic application in your own field.


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TOPIC RECAP

# Recap — Quantum Error Correction

- Why QEC is needed
- Quantum-specific challenges
- Syndrome measurement
- Bit-flip and phase-flip codes
- Shor, Steane and surface codes
- Fault tolerance


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WRAP-UP

# Course Summary & Next Steps


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LEARNING OUTCOMES ACHIEVED

# What You Achieved

1

## Quantum foundations

Built the state vectors, unitary and Hermitian operators behind every algorithm.

2

## Quantum systems

Visualised qubits on the Bloch sphere and ran circuits on real IBM hardware.

3

## Gates and circuits

Applied the full gate set and constructed Bell and GHZ entangled states.

4

## Quantum protocols

Implemented teleportation, superdense coding and BB84 with eavesdropper detection.

5

## Quantum algorithms

Ran Deutsch, Deutsch-Jozsa, Bernstein-Vazirani, Grover and the QFT.

6

## Cryptography and safety

Ran Shor's period finding and built a PQC migration plan.

7

## Applications

Simulated $H_2$ with VQE and optimised with QAOA.

8

## Error correction

Encoded, diagnosed and corrected errors with the bit-flip, Shor and Steane codes.


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NEXT STEPS

# Continue Your Quantum Journey

1. **IBM Quantum Learning**
   Free courses and challenges at learning.quantum.ibm.com.

2. **Qiskit documentation**
   The reference for every gate, primitive and runtime feature.

3. **PennyLane demos**
   Excellent tutorials on variational and quantum-machine-learning methods.

4. **NIST PQC**
   Track FIPS 203/204/205 adoption and vendor migration guidance.

5. **Practise the labs**
   Re-run every activity in this course from a blank notebook.

6. **Stay current**
   Quantum hardware roadmaps shift yearly — follow the vendor announcements.


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FINAL ASSESSMENT

# Assessment

## Written Assessment (WA)

- Short-Answer Questions (SAQ)
- 1 hour · open book
- Tests underpinning KNOWLEDGE
- Covers all eight topics

## Practical Performance (PP)

- Hands-on quantum tasks
- 1 hour · open book
- Tests PRACTICAL ability
- Built on the class activities

**OPEN BOOK — course slides, Learner Guide and approved materials only**

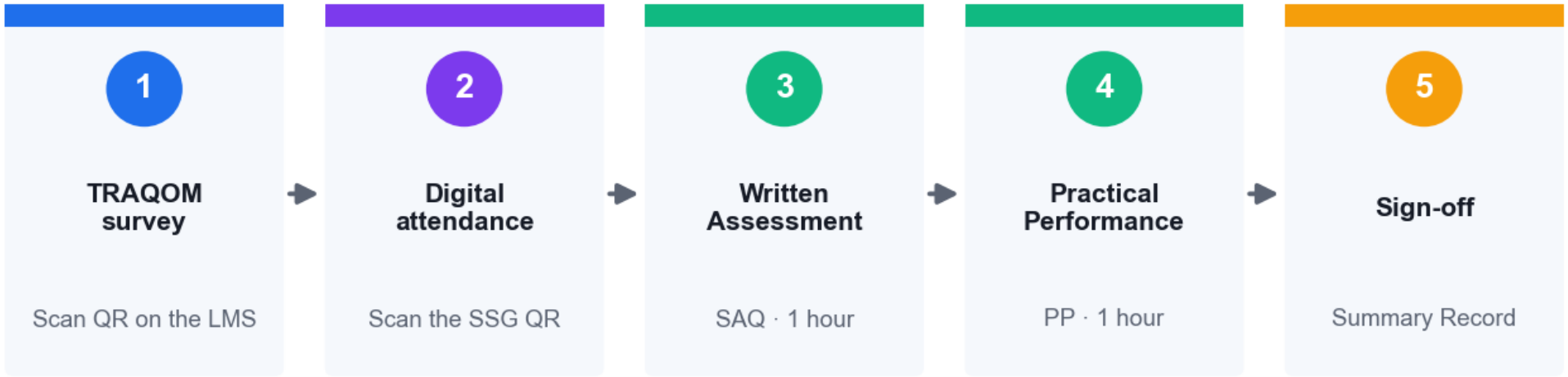
**Eligibility:** minimum 75% attendance + assessed Competent. An appeal process is available.


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ON ASSESSMENT DAY

# Assessment Flow

## What happens on assessment day




**Open book — course slides, Learner Guide and approved materials only**

Complete every step in order — the TRAQOM survey and digital attendance are mandatory for funding.


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# Digital Attendance (Mandatory)

1. Trainer displays the digital attendance QR code from the SSG portal
2. Scan the QR code with your mobile phone camera
3. Submit your attendance — AM, PM and Assessment, every day
4. 75% minimum attendance is required for assessment and funding


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END OF COURSE

# Thank You!

You can now design quantum models, apply quantum gates, select quantum algorithms and plan a post-quantum migration — the skills that secure the future.


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